SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics

COURSE OUTCOME COVERED

COURSE CODE: CSA15

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Question Count: 5

Code of Conduct

Total Marks: 25

I __**Shaik Mohammad Akhim(192525132)**_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student**: _Shaik Mohammad Akhim**_____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage.

How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

ANSWERS:

1. Auto-Scaling and Load Balancing During Seasonal Sales

Introduction:

Cloud auto-scaling and load balancing help companies manage sudden increases in website traffic during seasonal sales.

1. Auto-Scaling:

- Auto-scaling automatically increases or decreases server resources based on user demand.
- During high traffic, additional servers are added.
- When traffic decreases, extra servers are removed.
- It reduces unnecessary resource usage and cost.

2. Load Balancing:

- Load balancing distributes incoming user requests across multiple servers.
- It prevents a single server from becoming overloaded.
- It improves application performance and availability.

3. Reducing Downtime:

- Multiple servers handle user requests.
- If one server fails, traffic is redirected to other servers.
- Users can access the website without interruption.

4. Role of Monitoring Tools:

- Monitoring tools track CPU usage, memory, traffic and server performance.
- They identify problems and help trigger auto-scaling.
- Examples: Amazon CloudWatch and Azure Monitor.

Real-World Example:

Amazon experiences heavy traffic during Great Indian Festival sales. Auto-scaling adds more servers, while load balancing distributes customer requests. This prevents website crashes and provides a smooth shopping experience.

Conclusion: Auto-scaling and load balancing improve availability, performance and customer satisfaction while reducing costs.

2. Secure Cloud Storage and Data Backup

Introduction:

Cloud storage allows organizations to store large amounts of backup data securely without maintaining physical storage systems.

1. Data Durability:

- Cloud providers store data in reliable storage systems.
- Data is protected against hardware failures.
- Backup copies help prevent permanent data loss.

2. Data Redundancy:

- Multiple copies of data are stored in different locations.
- If one storage system fails, another copy can be used.
- It improves data availability.

3. Encryption:

- Encryption converts readable data into an unreadable format.
- It protects sensitive information from unauthorized users.
- Data is encrypted during storage and transfer.

4. Access Control:

- Only authorized users can access backup data.
- Permissions and authentication protect important files.
- Multi-factor authentication provides additional security.

5. Cost Advantages:

- No need to purchase expensive physical storage devices.
- Reduces maintenance and electricity costs.
- Pay-as-you-use pricing helps save money.

Disaster Recovery Example:

A hospital stores patient records in cloud storage. If its local server is damaged due to a fire, the hospital can restore patient records from cloud backups and continue its operations.

Conclusion: Cloud storage provides secure, reliable and cost-effective data backup with better disaster recovery.

3. Content Delivery Networks (CDNs) and Global Application Deployment

Introduction:

A Content Delivery Network (CDN) is a group of distributed servers that deliver content to users from locations closer to them.

1. CDN:

- CDN stores copies of website content on different servers.
- Users receive data from the nearest available server.
- It reduces network delay and improves loading speed.

2. Edge Servers:

- Edge servers are located closer to users.
- They process and deliver content quickly.
- They reduce the distance travelled by data.
- They improve video streaming and gaming performance.

3. Global Availability:

- Cloud providers maintain data centers in different geographical regions.
- Applications can be deployed across multiple regions.
- If one region fails, another region can provide services.

4. Performance Improvement:

- Reduces latency.
- Improves download speed.
- Handles large numbers of users.
- Provides a smooth user experience.

Real-World Example:

Netflix uses content delivery infrastructure to provide video streaming to users worldwide. Videos can be delivered from nearby servers, reducing buffering and improving streaming quality.

Conclusion: CDNs and edge servers provide faster content delivery, reduced latency and better global availability.

4. Cloud Security and Regulatory Compliance in Financial Institutions

Introduction:

Financial institutions handle sensitive customer information and transactions. Therefore, they require strong cloud security and regulatory compliance.

1. Regulatory Compliance:

- Cloud providers support security standards and compliance requirements.
- They provide tools for auditing and reporting.
- Organizations must ensure that their cloud configuration meets applicable regulations.

2. Logging:

- Logging records user activities and system events.
- It helps identify unauthorized access.
- It provides information for security investigations.

3. Monitoring:

- Monitoring continuously checks cloud resources.
- It detects unusual activities and possible threats.
- It helps administrators respond quickly.

4. Encryption:

- Encryption protects sensitive financial information.
- Data is encrypted during storage and transmission.
- Only authorized users can access the information using proper keys.

5. Data Sovereignty:

- Data sovereignty means storing and processing data according to the laws of a particular country.
- Organizations can select suitable cloud regions.
- They must ensure that data storage and transfer follow local regulations.

Real-World Example:

A bank stores customer account details and transaction records in the cloud. Encryption protects transaction information, while logging records every transaction. Access control prevents unauthorized users from accessing customer accounts.

Conclusion: Cloud security features help financial institutions protect sensitive data, maintain compliance and build customer trust.

5. Containerization and Kubernetes in Cloud Deployment

Introduction:

Containerization is a technology that packages an application and its required libraries into a single unit called a container.

1. Containers vs Virtual Machines:

Containers	Virtual Machines
Share the host operating system kernel.	Have separate guest operating systems.
Lightweight.	More resource-intensive.
Start quickly.	Take more time to start.
Use fewer resources.	Use more resources.
Easy to deploy.	Require more resources to manage.

2. Container Orchestration:

- Container orchestration manages multiple containers.
- Kubernetes is a popular container orchestration platform.
- It automates deployment and management.
- It automatically restarts failed containers.
- It supports scaling based on application demand.

3. Portability:

- Containers can run on different environments.
- They support deployment across different cloud providers.
- They reduce compatibility problems.
- They make application development easier.

4. Microservices Application:

- A microservices application is divided into small independent services.
- Each service performs a specific function.
- Containers allow each service to run separately.
- Kubernetes manages and scales these services.

Real-World Example:

An online shopping application contains different services such as:

- User login service.
- Product management service.
- Payment service.
- Order management service.
- Delivery tracking service.

Each service runs in a separate container. Kubernetes manages these containers and increases their number when user traffic increases.

Conclusion: Containerization and Kubernetes improve application portability, scalability, resource utilization and deployment efficiency.

Quick Revision – Important Points

Question	**Main concepts**
1. Seasonal sales	Auto-scaling, load balancing, monitoring
2. Cloud backup	Durability, redundancy, encryption, access control
3. Global deployment	CDN, edge servers, latency, availability
4. Financial security	Compliance, logging, monitoring, encryption
5. Containerization	Containers, VMs, Kubernetes, microservices

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight

Completeness	15%	Response covers all required	Covers most aspects	Covers only basic aspects	Incomplete response
		aspects			
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity